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From Mature Production Wells to Clean Energy: Temperature Calibration for Offshore Geothermal Potential

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Abstract

As part of a broader initiative to explore geothermal energy as a means to electrify offshore assets in the Malay Basin and reduce carbon emissions, the prefeasibility study was performed to evaluate the viability of repurposing mature oil fields for geothermal heat extraction. The study focuses on improving the accuracy of formation temperature estimates, a critical input for geothermal resource assessment, by correcting legacy downhole temperature measurements from a representative offshore field ("Field A").

Historical temperature data, originally collected during oil and gas operations, exhibit significant uncertainties due to cooling effects from drilling mud circulation, gas expansion, and delayed measurement timing. This study applies a multi-method correction workflow to two key deep wells ("Well A1" and "Well A2") that penetrate the target geothermal zones ("Zone A," "Zone B," "Zone C"). The approach includes cross-well validation with newer offset wells, a field-specific Horner-type time-log correction model, and adjustments for Joule–Thomson cooling in gas-bearing intervals. Corrections of +24 °C and +14 °C were applied to Well A1 and Well A2, respectively.

Corrected geothermal gradients revealed higher formation temperatures than previously assumed, with updated thermal ranges across the target zones as follows: Zone A: 140–170 °C, Zone B: 158–185 °C, and Zone C: 180–230 °C. These revised temperature profiles were used as initial input conditions for static and dynamic geothermal reservoir simulation. The modeling aimed to estimate long-term heat production rates, evaluate sustainability of energy output, and optimize injection–production strategies (doublet injection–production designs, etc.) under various development scenarios. The results demonstrated that re-evaluation of legacy temperature data is essential to reduce thermal uncertainty and to produce reliable forecasts for energy recovery potential in geothermal developments.

This study introduces a practical, field-validated methodology for correcting historical oilfield temperature data, enabling more accurate geothermal gradient determination and improved thermal model reliability. By addressing the bias introduced by outdated or incomplete measurements, the workflow enhances the accuracy of heat-in-place calculations and reservoir simulation outcomes. The correction strategy serves as a critical enabler for identifying high-potential geothermal zones and supports the broader

offshore decarbonization roadmap by unlocking cleaner energy solutions through repurposing existing hydrocarbon infrastructure.

Overall, this work contributes a replicable framework for offshore geothermal prefeasibility screening in mature basins, promoting more confident investment decisions and supporting the energy transition objectives of hydrocarbon producers in Southeast Asia and beyond.

Objectives

A critical barrier in evaluating offshore geothermal potential within mature oil and gas fields is the reliability of legacy downhole temperature data. Historically, these measurements were acquired during hydrocarbon drilling operations where formation temperature was not a primary objective. As a result, they are often underestimated due to cooling effects from mud circulation, gas expansion, and timing of acquisition. Accurate temperature calibration is therefore a foundational step in assessing the true geothermal gradient and screening subsurface heat potential.

The primary objective of this study is to validate and correct historical temperature measurements from selected offshore wells in the Malay Basin to enable more accurate estimation of geothermal gradients and reduce thermal model uncertainty. The focus is on applying field-calibrated correction techniques, including circulation-time corrections and cross-well validation, to enhance the reliability of temperature profiles across geothermal target zones.

These corrected profiles directly support key phases of the geothermal feasibility assessment, including:

- Screening and mapping of high-potential geothermal zones, enabled by improved thermal characterization and enhanced integration of corrected temperature data with regional and field-scale geological frameworks;
- Integration into basin-scale Petroleum System Models (PSM) and integrate with Forward Stratigraphic Models (FSM) to establish consistent thermal models across scales, supporting more accurate identification of geothermal hotspots, reservoir extension and quality distribution;
- Input into thermal and dynamic thermal simulation models to forecast long-term heat production capacity, temperature sustainability, and fluid behavior under various injection–production scenarios;
- Optimization of development strategies for offshore electrification, ensuring alignment with decarbonization goals while minimizing subsurface uncertainties and maximizing energy recovery.

While the broader feasibility study (Fig. 1) includes regional thermal modeling, petroleum system integration, and reservoir simulation, this paper emphasizes temperature correction as the enabling step that underpins the credibility and success of downstream geothermal evaluation and planning.

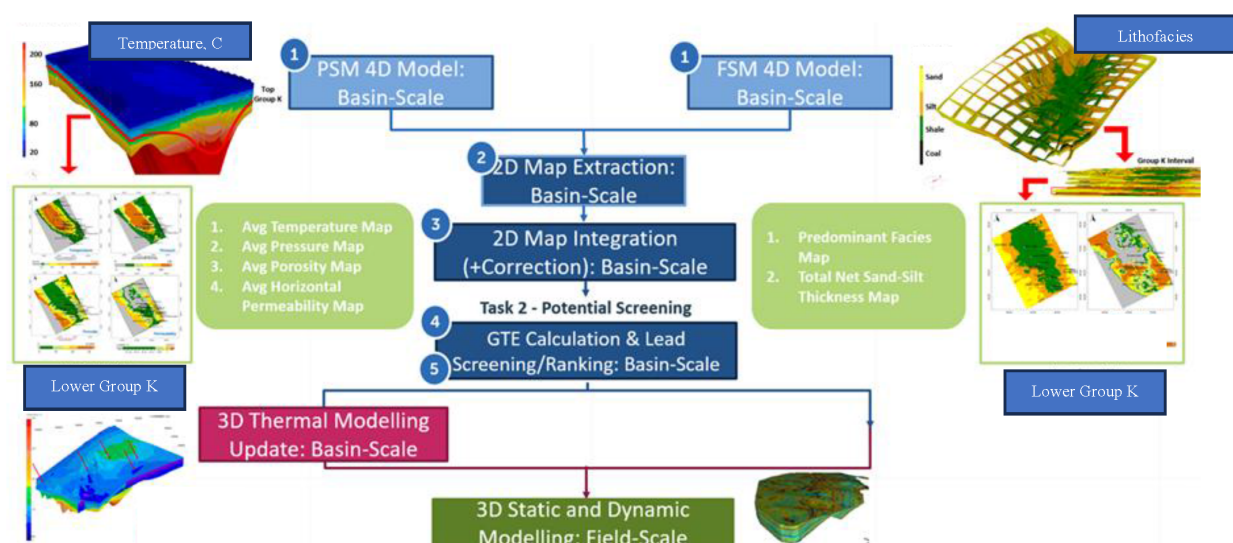


Figure 1—Broader Geothermal Feasibility Study Workflow (Source: Generated by Authors)

Field Temperature data, limitations and geothermal gradient

This study focuses on a key offshore field within the Malay Basin identified as having a potentially high geothermal gradient, making it a strong candidate for geothermal energy development. Accurate characterization of the subsurface temperature profile in this field is essential to evaluate its geothermal potential.

A total of 31 wells are available in the field area, with various forms of temperature and pressure data distributed as follows:

- 11 wells with Modular Formation Dynamics Tester (MDT) data,
- 3 wells with Repeat Formation Tester (RFT) data,
- 15 wells with Static Gradient Survey (SGS) temperature readings,
- Several wells with Drill Stem Test (DST) data, including both gas and liquid tests

Among these, the most accurate and reliable temperature measurements are obtained from liquid DSTs, which are used as calibration benchmarks. RFT data are considered the least reliable, while MDT and gas-phase DST data require correction. These corrections are based on:

- Circulation time for MDT readings, using a field-derived Horner-type model, and
- Pressure drawdown for gas DST intervals to account for Joule–Thomson cooling.

There are two most critical deep wells - "Well A1" and "Well A2" -penetrate into the deeper geothermal target zones ("Zone A," "Zone B," "Zone C"). These wells serve as reference wells for geothermal evaluation within the target geothermal stratigraphic intervals.

Correcting and harmonizing the temperature data from these wells provides the foundation for building an accurate geothermal gradient model and enables more confident resource screening and simulation across the field.

The geothermal gradient in the Field has been analyzed using the data before temperature correction (Fig.2).

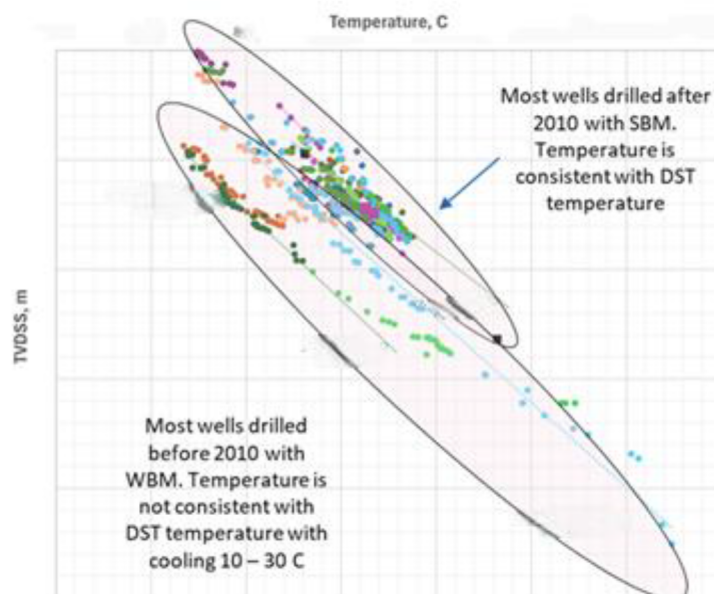


Figure 2—Field Temperature Data plot (temperature vs. depth (SSTVD)) (Source: Generated by Authors)

Two distinct clusters of geothermal trends are evident:

1. Pre-2010 wells (primarily appraisal and exploration wells) characterized by MDT measurements that consistently report lower formation temperatures;
2. Post-2010 wells and DST measurements, which record systematically higher temperatures-by approximately 10 to 30 °C-compared to nearby older wells.

The discrepancy can be explained by multiple factors:

- Mud type effects: Earlier wells were drilled using Water-Based Mud (WBM), which retains and dissipates heat, leading to underestimated temperature readings. In contrast, more recent wells used Synthetic-Based Mud (SBM), which results in reduced cooling and better preservation of formation temperature. In some wells WBM is applied for shallow sections and SBM for deeper sections and there is temperature trend increase for deeper sections below the depth when SBM are used.
- Time since circulation: Temperature measurements taken shortly after mud circulation are prone to significant cooling errors. For instance, in one of the exploration well, a comparison of multiple time-lapse measurements was performed showing a temperature deviation of up to 38 °C, underscoring the importance of accounting for thermal recovery time.

Field Temperature data corrections and benchmarks

The most reliable temperature measurements are derived from liquid DSTs (oil/water intervals), which are considered the primary benchmark for calibration due to minimal disturbance from cooling effects. These high-confidence data points are used to correct the well and nearby wells temperature sources across the field (Mohammadreza Zare-Reisabadi, M.R. Kamali, M. Mohammadnia, F. Shabani).

Temperature data sources subjected to correction include:

- MDT/RFT measurements from wells drilled with SBM and characterized by short circulation times. These values are calibrated against nearby WBM wells and/or liquid DST benchmarks to account for potential underestimation.

- DST measurements in gas zones, which are adjusted for the Joule–Thomson cooling effect using pressure differential-based correction.

In gas reservoirs, downhole temperatures measured during Drill Stem Tests (DSTs) can be lower than true formation temperatures due to the Joule–Thomson cooling effect. This phenomenon occurs when gas flows from high-pressure reservoir conditions to a lower-pressure wellbore environment during the test. The gas undergoes adiabatic expansion, resulting in a temperature drop that must be accounted for when interpreting temperature logs. To correct for this cooling to estimate the expected temperature drop based on the pressure differential:

$$\Delta T = \mu_{JT} \cdot (P_{res} - P_{wf}) \quad (\text{Eq. 1})$$

where:

- ΔT is the temperature correction in °C (cooling effect),
- μ_{JT} is the Joule–Thomson coefficient in °C/bar, based on gas composition and conditions, and is best derived from PVT analysis
- $(P_{res} - P_{wf})$ is the pressure difference between reservoir pressure (P_{res}) and flowing bottomhole pressure (P_{wf}) during DST, in bar.

The corrected MDT temperature profiles from two deep wells - the only wells penetrating the target geothermal zones-serve as the primary inputs for geothermal modelling.

Field Temperature correction using Horner equation

A key factor contributing to underestimated downhole temperatures is the cooling effect induced by drilling mud circulation. In the Field, the time since circulation (TSC) prior to temperature logging varies between approximately 8 and 35 hours (Fig. 3). To quantify this effect and calibrate measured temperatures, a field-specific correction model was developed based on temperature data from one of the well.

SUITE #	RUN #	MAX DEPTH M	TIME SINCE CIRC. STOPPED Hrs: Mins	THERMOMETERS		
				Deg C	Deg C	Deg C
1	1	875	08:45	62.2	62.2	63.3
2	1	1764	9:15	93.3	91.7	91.1
2	2	1763	19:20	104.7	104.7	104.7
2	3	1761	N/A			
2	4	1773	N/A			

Time since circulation stop	Cooling effect, C
5 hours	37.3
24 hours	11.2
35 hours	5.5

Figure 3—Well Temperature Data (t vs. time since circulation) (Source: Generated by Authors)

A key factor contributing to the systematic underestimation of downhole formation temperatures in historical datasets is the cooling effect induced by drilling mud circulation during drilling operations. The extent of this cooling is strongly influenced by the time since circulation (TSC - the elapsed time between stopping drilling fluid circulation and recording temperature measurements. In the case of the studied

offshore field, TSC values recorded across wells vary significantly, ranging from as short as 8 hours to as long as 35 hours prior to temperature acquisition (Fig. 3). These relatively short stabilization periods are often insufficient for the wellbore.

To quantify and correct this cooling effect, a field-specific temperature correction model was developed using temperature data from one of the appraisal wells. This well contains multiple downhole temperature measurements captured at the same depth but at different TSC intervals, thereby enabling empirical analysis of thermal recovery trends. The data show that between 9 and 19 hours after stopping circulation, the measured temperature increased by approximately 13.6 °C. This clearly indicates that significant thermal re-equilibration was still ongoing during this timeframe, and that the initial temperature readings did not represent true formation conditions.

This observation supports the application of a Horner-type extrapolation to estimate the true formation temperature by correcting for the remaining thermal lag (Douglas Waples, May Ritt Pedersen). The derived field-specific correction equation for the Field is:

$$\Delta T = 51.2 \times e^{-0.0635 \times tsc} \quad (\text{Eq. 2})$$

where:

- ΔT is the temperature correction in °C (cooling effect),
- tsc is the time since circulation stopped, in hours.

This Eq. 2 enables the adjustment of measured temperatures to reflect a more accurate approximation of the undisturbed formation temperature. The shape and behavior of the derived equation are consistent with global mud-logging and Modular Dynamic Tester (MDT) correction libraries, which recommend stabilization periods of at least 50 hours to reduce temperature uncertainty to within ± 5 °C. However, in real-world operations-especially in older wells-such extended waiting times are rarely achieved due to operational constraints.

By leveraging this locally calibrated correction model, temperature measurements from MDT or RFT tools conducted under suboptimal TSC conditions can now be retroactively corrected. This significantly improves the reliability of geothermal gradient estimation, particularly for key reference wells that intersect deep prospective intervals. The correction function plays a pivotal role in upgrading the temperature dataset used for geothermal screening, reservoir modeling, and energy yield estimation across the field, and ultimately supports better-informed decisions regarding the potential for geothermal energy development in legacy hydrocarbon assets.

Deep Well A1 temperature correction

Well A1 was drilled before 2010 as part of early appraisal activities in the field. The well was completed using a combination of drilling fluids: water-based mud (WBM) was utilized for the upper intervals (shallower than about 2500 m TVDSS), while synthetic-based mud (SBM) was employed in the deeper sections (greater than 2500 m TVDSS). This drilling mud transition is significant because WBM is known to have a higher thermal conductivity and lower heat retention, which often leads to stronger cooling effects and underestimation of formation temperatures in conventional wireline measurements.

As illustrated in Fig. 4, two distinct geothermal trends are observed, with a noticeable shift in the temperature profile occurring at approximately 2500 TVDSS.

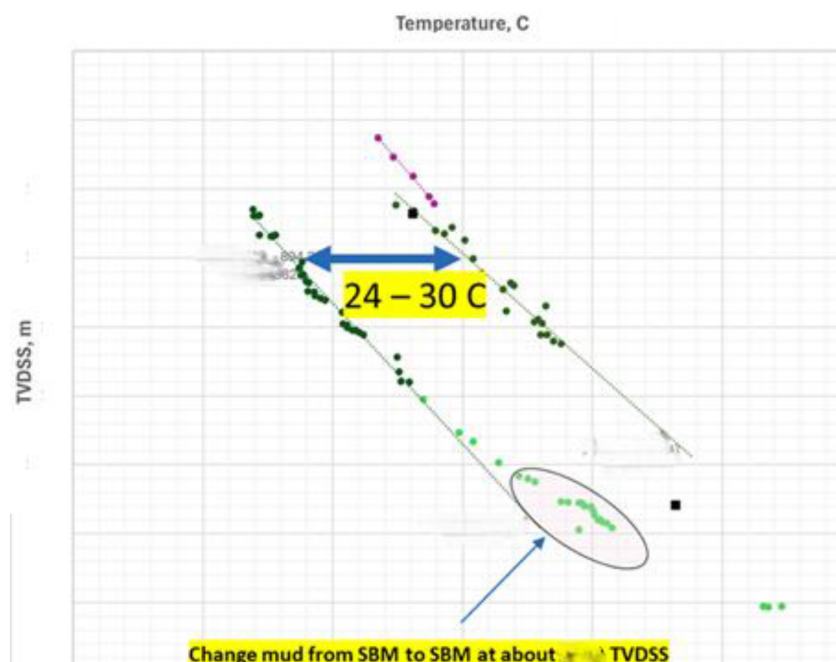


Figure 4—Well A1 Temperature Correction (Source: Generated by Authors)

Well A1 is located approximately 650 meters from a nearby producer well, which was drilled post-2010. Despite this proximity, the formation temperatures measured in Well A1 are systematically 24 to 30 °C lower than those recorded in producer well at equivalent depths. This offset is evident both in the shallow section (about <2500 m TVDSS), where a 30 °C difference is observed, and in the deeper section (about >2500 m TVDSS), where the discrepancy narrows to approximately 24 °C (Fig 4.).

The temperature trend in Well A1 for depths shallower than about 2500 m aligns with pre 2010 drilled wells, while the producer temperature profile is consistent with post-2010 drilled wells, as well as with DST-derived temperature data. This systematic offset-24 °C at depths greater than about 2500 m and up to 30 °C at shallower intervals-has also been confirmed by Well A1 DST data, which closely matches post 2010 wells temperature profiles. Based on this consistent offset, a correction of +24 °C is applied to Well A1 temperature data to align it with the more reliable reference wells and DST benchmarks.

Deep Well A2 temperature correction

Deep Well A2, located approximately 1 km from Well A1, was drilled in 2014 using synthetic-based mud (SBM), which generally helps to mitigate cooling effects commonly associated with traditional water-based muds (WBM). Despite the use of SBM, temperature measurements acquired via Modular Formation Dynamics Tester (MDT) in Well A2 were consistently found to be 14–16 °C lower than those recorded in nearby producer wells drilled after 2010.

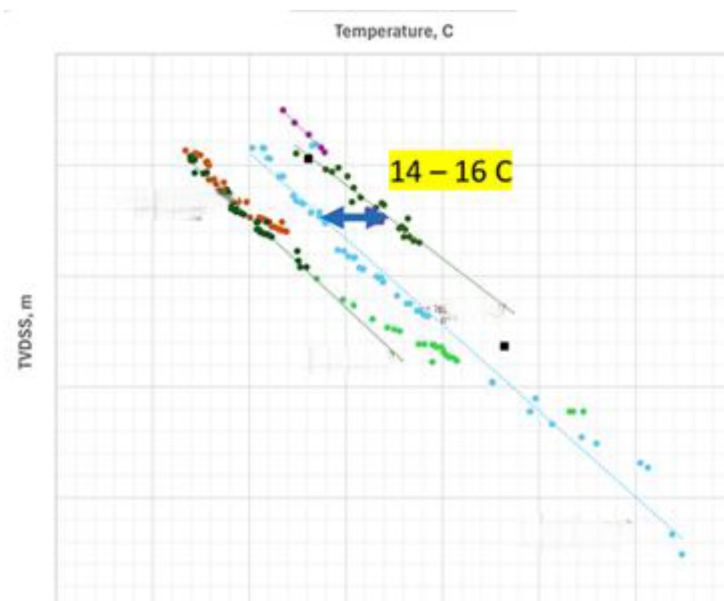


Figure 5—Well A2 Temperature Correction (Source: Generated by Authors)

This systematic offset is observed throughout the entire measured depth interval, suggesting that cooling effects persisted even under improved mud conditions—likely due to shorter time since circulation (TSC) prior to logging and possible localized mud losses or thermal disturbances during drilling. DST temperature data from Well A1 confirms temperature measurements performed in producers wells. Thus Well A2 temperature was corrected by +14.1 °C.

Field Geothermal Gradient and Uncertainty Evaluation

Fig. 6 presents a compilation of subsurface temperature data across the Field, including corrections applied to key wells that penetrate the target geothermal intervals—namely Zones A, B, and C. Well A1 and Well A2 are the only wells penetrating target geothermal zones. Temperature measurements from these wells were calibrated based on nearby reference wells and DST benchmarks. A correction of +24 °C was applied to Well A1, while Well A2: +14 °C correction to align with post-2010 temperature trends and validated DST data. In addition, DST measurements acquired during gas production tests—which are susceptible to Joule–Thomson cooling effects—were corrected by an estimated 7 to 10 °C, depending on the pressure differential observed in the tested interval. These adjustments ensure a more accurate reflection of in-situ formation temperatures and reduce the uncertainty introduced by transient cooling effects during well testing.

Five wells were excluded from the main temperature plot due to their limited depth of penetration (not reaching the target geothermal zones) and clear evidence of cooling effects that would otherwise distort the geothermal gradient assessment. For the remaining 17 wells, original temperature measurements were retained, including the unadjusted liquid-phase DST temperature.

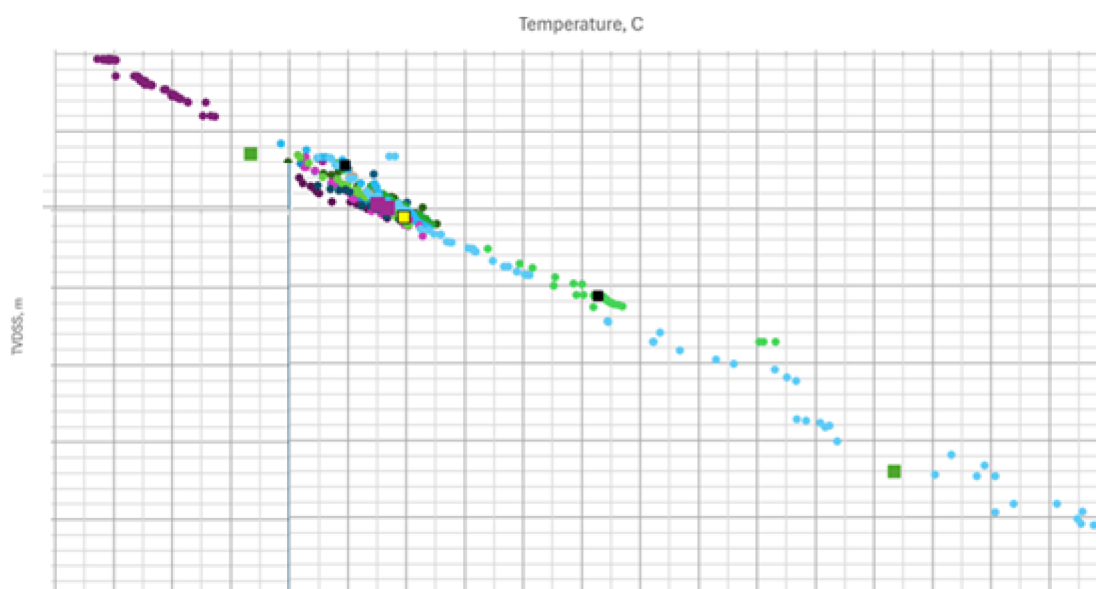


Figure 6—Field Temperature Data plot (temperature vs. depth (SSTVD)) (Source: Generated by Authors)

Corrected geothermal gradients revealed significantly higher formation temperatures in the target zones than previously estimated. Updated temperature ranges are:

- Zone A: 140–170 °C
- Zone B: 158–185 °C
- Zone C: 180–230 °C

These corrected temperature profiles were used in geothermal reservoir simulation to evaluate heat production rates and optimize injection–production design for sustainable power generation. The results confirmed the need to reprocess legacy temperature data.

To capture temperature-related uncertainty in geothermal modeling, a scenario-based approach was implemented. Three cases were defined to take into account the uncertainty range of temperature within the target reservoir:

- Base Case: Reflects corrected temperatures, aligned with post 2010 wells and liquid DST benchmarks.
- Low Case: Represents a conservative scenario approximating uncorrected data, defined as Base -15 °C.
- High Case: Accounts for potential underestimation, defined as Base $+10$ °C, consistent with the upper boundary of observed temperature data.

These temperature cases were used as initial conditions in the geothermal reservoir model to evaluate heat-in-place, thermal sustainability, and development feasibility under varying subsurface conditions.

Conclusions and Recommendations

This study demonstrates that temperature calibration is an essential first step in unlocking the geothermal potential of mature offshore oil and gas fields (Erin Camp, Teresa Jordan). The quality and accuracy of subsurface temperature data significantly influence decisions in a geothermal development workflow - from assessing geothermal gradients to estimating heat-in-place, reservoir productivity, and ultimately, power generation potential.

By applying systematic corrections to historical temperature measurements - such as DSTs and BHTs affected by drilling-induced cooling - we were able to reconstruct a more reliable thermal profile for the Malay Basin. These corrected values were then integrated into a basin-scale Petroleum System Model (PSM), resulting in an updated view of present-day subsurface temperature distribution, thermal gradients, and reservoir conditions.

The recalibrated model enabled us to identify high-potential geothermal prospects. The field described in this paper was shortlisted and identified as a particularly promising candidate due to its relatively high geothermal gradient, deep relatively good rock quality reservoirs, and two existing deep well penetrating the target geothermal zone. The combination of high temperature, substantial net sand thickness, adequate porosity and permeability, and elevated reservoir pressure positions the Field as a viable target for geothermal energy production using either single-well or doublet concepts.

Ultimately, this work highlights that geothermal repurposing of offshore brownfields is technically feasible (Andres Negre, Khadija Andjar, et al.) and potentially impactful, provided that input data - particularly subsurface temperature - is managed with the same technical precision as in oil and gas workflows. In conclusion, temperature correction is critical step in geothermal feasibility studies. It supports better hotspot identification, reservoir characterization, and development planning. The findings of this study provide a practical framework for applying this methodology to other offshore basins across Southeast Asia and globally. As the energy industry transitions toward low-carbon solutions, the intelligent reuse of mature assets such as the Field - guided by high-integrity temperature data - offers a realistic and sustainable pathway to clean energy generation.

Nomenclature

Acronyms

BHT	–	bottomhole temperature
DST	–	Drill Stem Test
FSM	–	Forward Stratigraphic Model
MDT	–	Modular Dynamics Test
PSM	–	Petroleum System Model
RFT	–	Repeat Formation Test
SBM	–	synthetic-based mud
SGS	–	Static Gradient Survey
STB	–	stock-tank barrel
SSTVD	–	subsea true vertical depth
TSC	–	time since circulation
TVDSS	–	true vertical depth subsea
WBM	–	water-based mud

Symbols

ΔT	–	temperature correction [$^{\circ}\text{C}$]
μJT	–	Joule–Thomson coefficient [$^{\circ}\text{C}/\text{bar}$]
Pres	–	reservoir pressure [bar]
Pwf	–	flowing bottomhole pressure [bar]
Tsc	–	time since circulation stopped [hours]

Units

$^{\circ}\text{C}$	–	degrees Celsius
h	–	hour

km – kilometer

m – meter

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